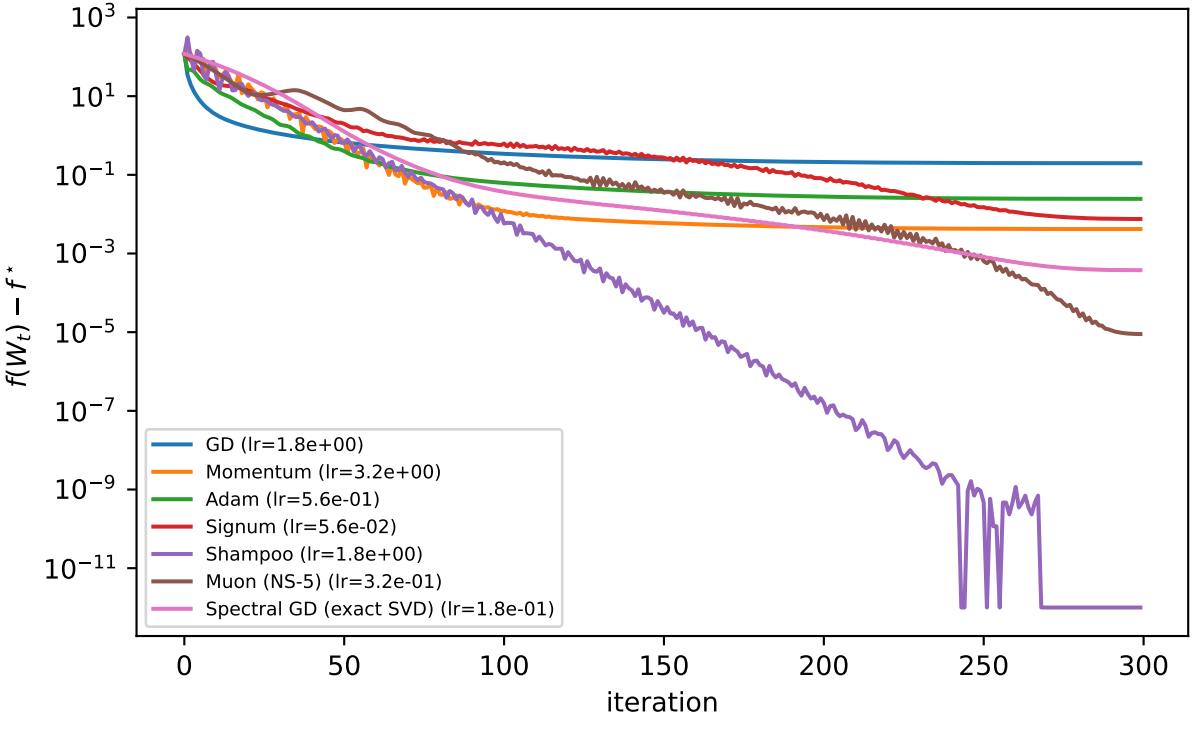
Matrix least squares, cond(X^T X)=1e4
$f(W_t) - f^\star$
iteration
10^{3}
10^{1}
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
0
50
100
150
200
250
300
GD (lr=1.8e+00)
Momentum (lr=3.2e+00)
Adam (lr=5.6e-01)
Signum (lr=5.6e-02)
Shampoo (lr=1.8e+00)
Muon (NS-5) (lr=3.2e-01)
Spectral GD (exact SVD) (lr=1.8e-01)